VANISHING OF H^1 OF A CURVE FIBRE FOLLOWS FROM GLOBAL R^1 VANISHING, OR AFTER ANY BASE CHANGE HITTING THE POINT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 20917-20928 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 20930-20940, the complete author proof (9 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Companion to the already-collected local-to-global pair L16078/L16085 in this lane; the 'or more generally after a base change hitting y' strengthening is the point of the card.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Morphisms, lemma openness-bounded-dimension-fibres; Limits, lemmas proper-top-cohomology-finite-type and higher-direct-images-zero-above-dimension-fibre (finite generation and base-change commutation of the top direct image); formal properties of fibres of quasi-coherent sheaves.

Source reference. Original lemma 0E7D: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 20940–20940. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0EX4: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 20940–20940. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02FZ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 20940–20940. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a proper morphism of schemes. Let $y \in Y$ be a point with $\dim(X_y) \leq 1$. If*

(1) *$R^1 f_* \mathcal{O}_X = 0$, or more generally*
(2) *there is a morphism $g : Y' \to Y$ such that y is in the image of g and such that $R^1 f'_* \mathcal{O}_{X'} = 0$ where $f' : X' \to Y'$ is the base change of f by g.*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Then $H^1(X_y, \mathcal{O}_{X_y}) = 0$.

Proof. To prove the lemma we may replace Y by an open neighbourhood of y. Thus we may assume Y is affine and that all fibres of f have dimension ≤ 1, see Morphisms, Lemma 02FZ. In this case $R^1 f_* \mathcal{O}_X$ is a quasi-coherent $\mathcal{O}_Y$-module of finite type and its formation commutes with arbitrary base change, see Limits, Lemmas 0EX4 and 0E7D. The lemma follows immediately. □